



AFRL

AEPW-4 HIGH-SPEED WORKING GROUP

AN OVERVIEW OF RECENT PROGRESS AND FUTURE DIRECTIONS

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AIAA AVIATION – AEROELASTIC PREDICTION WORKSHOP

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HSWG Key Questions

Objective: Assess the SoA of aerothermoelastic toolsets in high-speed applications

- What are the physical mechanisms that drive the various types of aerothermoelastic instabilities in high-speed flows?
- How accurately can dynamic aerothermoelastic instabilities be calculated? (Identifying onset of the instability vs the post-threshold behavior)
- Develop guidelines/metrics for modeling instabilities: What level of model fidelity is required? How much accuracy is lost when using lower fidelity methods?
- What is the uncertainty in our models? How does uncertainty propagate when coupling multiple models?
- What are the gaps/uncertainties in current experimental datasets that need to be addressed with follow-on or new experiments?
- How well do the SoA models handle complex structures and flow environments (transition, separation, SBLI, 3-D effects)?



HSWG Update Overview & Progress Towards Objectives

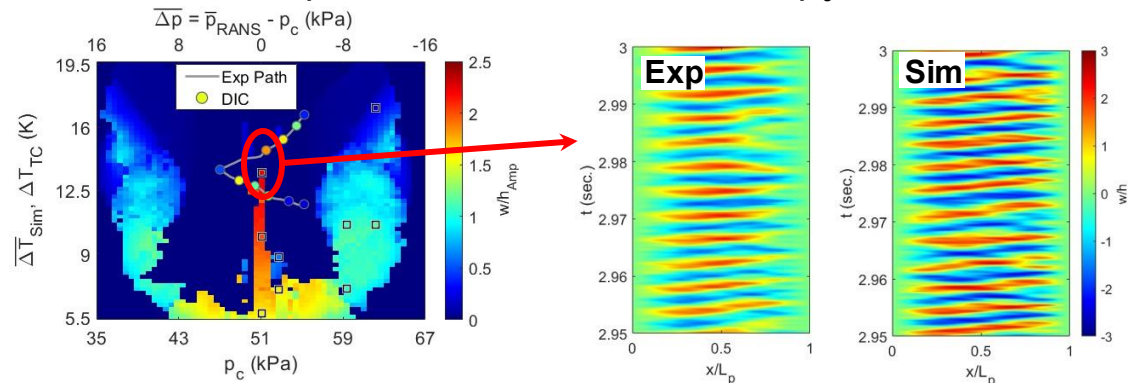
- Selected challenge problems:
 - RC-19: Large-amplitude, nonlinear dynamics of a thin panel with and without SBLI
 - HyMAX: Linear response of a cantilevered plate to transitional, separated SBLI
- HSWG off-cycle relative to other working groups under AePW
 - First formal workshop held at SciTech 2023 (informal meetup in 2024)
 - Monthly meetings throughout 2024/2025 highlighting participants' progress
- Current participation: 109 members on the email chain, 8 groups working on RC-19, 5 groups working on HyMAX
- **Near term:** Wrap up current iteration by SciTech 2026 (presentation of results/lessons learned)
- **Long term:** Selection of follow-on challenge problem

	AFRL-SSC	Duke	NASA	DLR	UNSW	MIT	Stevens	UC/ARL	Metacomp	Hexagon
RC-19	✓	✓	✓	✓	✓			✓	✓	✓
HyMAX		✓			✓	✓	✓	✓		

HSWG Participant Highlights/Results: RC-19 Without SBFI

AFRL Fluid + Structural ROMs

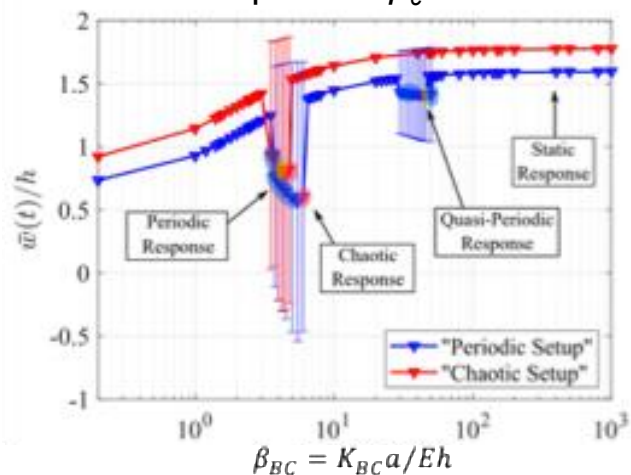
Coupled simulations with variable p_c and ΔT



Brouwer et al., ND 2021

Duke Fluid + Structural ROMs

Specified p_c and ΔT with variable stiffness BC



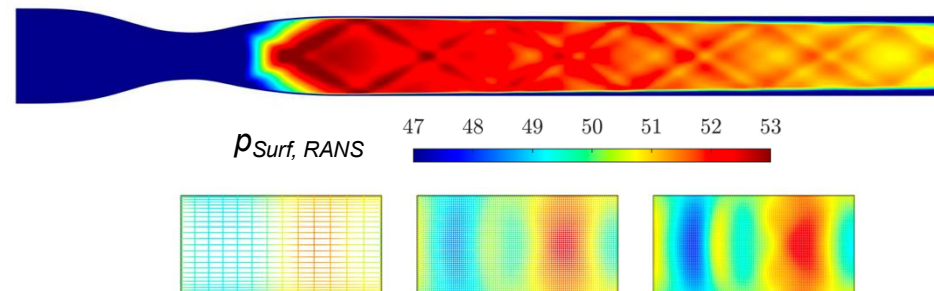
	Periodic Parameters	Chaotic Parameters
Δp (kPa)	3.91	Δp (kPa) 5.01
ΔT (K)	12.8	ΔT (K) 14.7

Boundary support parameter (β_{BC}) determined from benchtop vibration test or FEA

Piccolo Serafim et al., JFS 2023

NASA FUN3D (and PT) + FEA

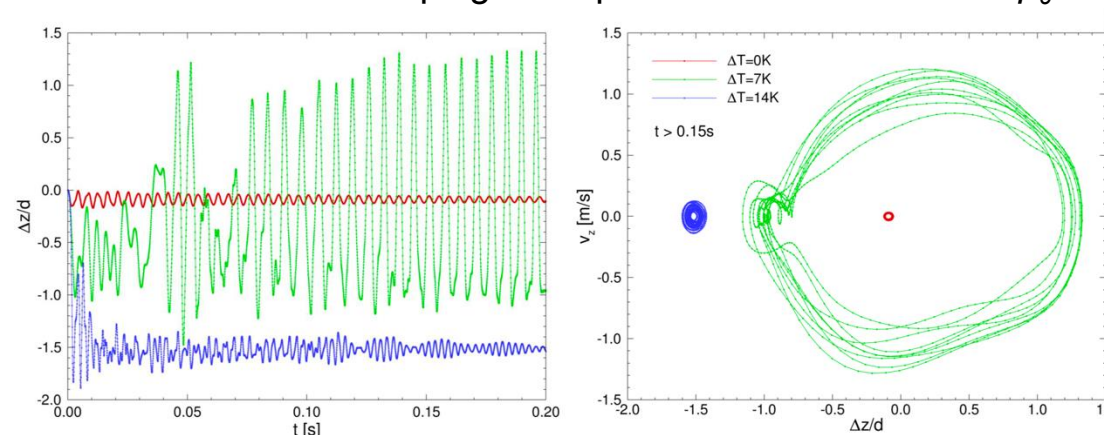
Rigid simulations of test section prior to coupled simulations



Stanford, AePW HSWG 2024

DLR URANS + FEA

No structural damping with specified ΔT and modeled p_c

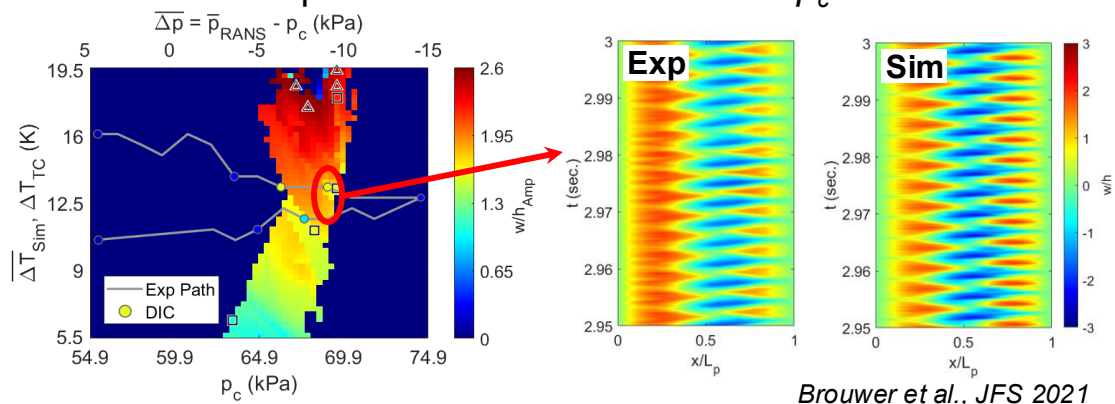


Reiman, AePW HSWG 2024

HSWG Participant Highlights/Results: RC-19 with SBLI

AFRL Fluid + Structural ROMs

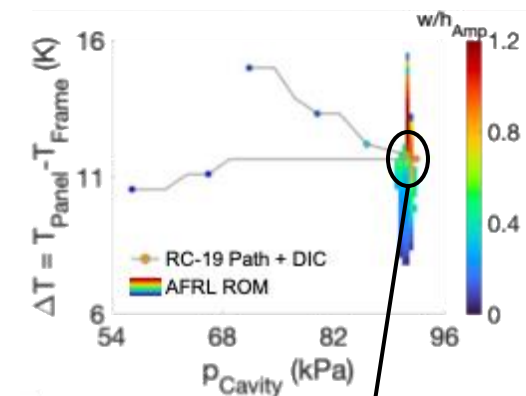
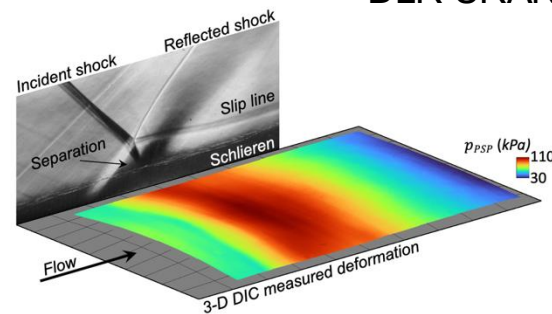
Coupled simulations with variable p_c and ΔT



Brouwer et al., JFS 2021

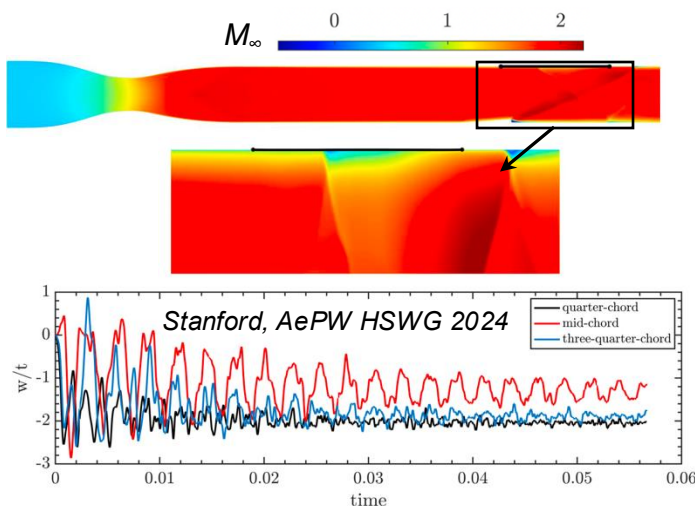
Separated SBLI (Not Officially Part of AePW)

AFRL Fluid + Structural ROMs
DLR URANS + FEA



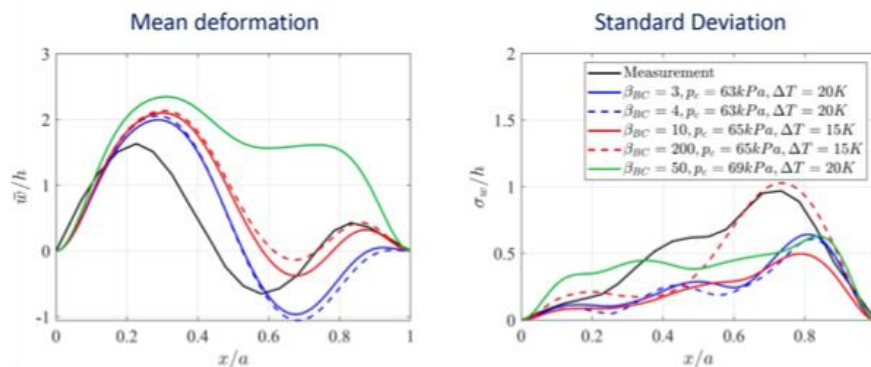
NASA FUN3D (and PT) + FEA

Coupled simulations with specified p_c and ΔT

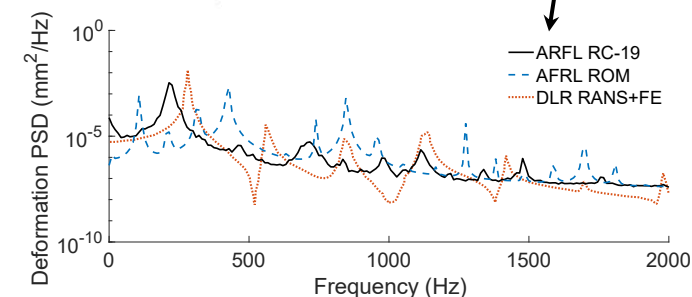


Duke Fluid + Structural ROMs

Variable p_c , ΔT , and stiffness BC



Piccolo Serafim et al., AIAA Aviation 2025



Reimann, AePW HSWG 2024



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HSWG Lessons Learned

- Low fidelity toolsets (Enriched piston & potential flow theories + structural ROMs) offer good balance of efficiency and predictive accuracy
 - Accurately predict onset of instabilities, but do not always accurately capture post-threshold response
 - Computational efficiency enables exploration of vast parameter space and more detailed mapping of instabilities
 - Tend to require less data from experiments to setup/initialize
- High-fidelity simulations (CFD + FEA)
 - Require additional data to setup/initialize – especially CFD (e.g., inflow BC, nozzle/test section geometry)
 - Computational expense limits computed time histories (problem for long-duration FSI phenomena)
 - Higher fidelity likely needed for separated SBLI cases – particularly if flow unsteadiness plays a role
 - Computational expense limits the number of parameter combinations – complicates direct comparison of simulations and experiments given input parameter uncertainties
- Limited use of data-driven multi-fidelity methods to tackle these challenge problems
- Additional needs from experiments:
 - Temperature field characterization (especially for buckled panels)
 - Improved characterization of configuration setup
 - Structural boundary conditions after install which can alter frequency characteristics
 - Test section inflow conditions
 - Improved baseline (no structural deformation) flow characterization for fluid model validation



HSWG Future Directions

- AFRL-Supported AE/ATE Experiments (Packaged consistent with RC-19 challenge problem):
 - RC-19 updates: Separated SBLI with snap-through & swept, attached SBLI with multiple instabilities
 - M6HRF: Compliant panel tests with quasi-static and dynamic responses (Led by Zach Riley)
 - H2K: Separated (transitional/turbulent) SBLI-induced aeroelastic experiments (Collaboration with DLR)
- Variations of HyMAX
 - Plans to test a similar configuration to HyMAX in the AFRL M6HRF
 - Will allow for longer flow times, $O(min)$, with the potential to observe flutter in the presence of thermal effects
- Other experiments/Inputs from AePW HSWG participants/AIAA FSI DG?



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HSWG Future Directions: AFRL M6HRF Challenge Problem

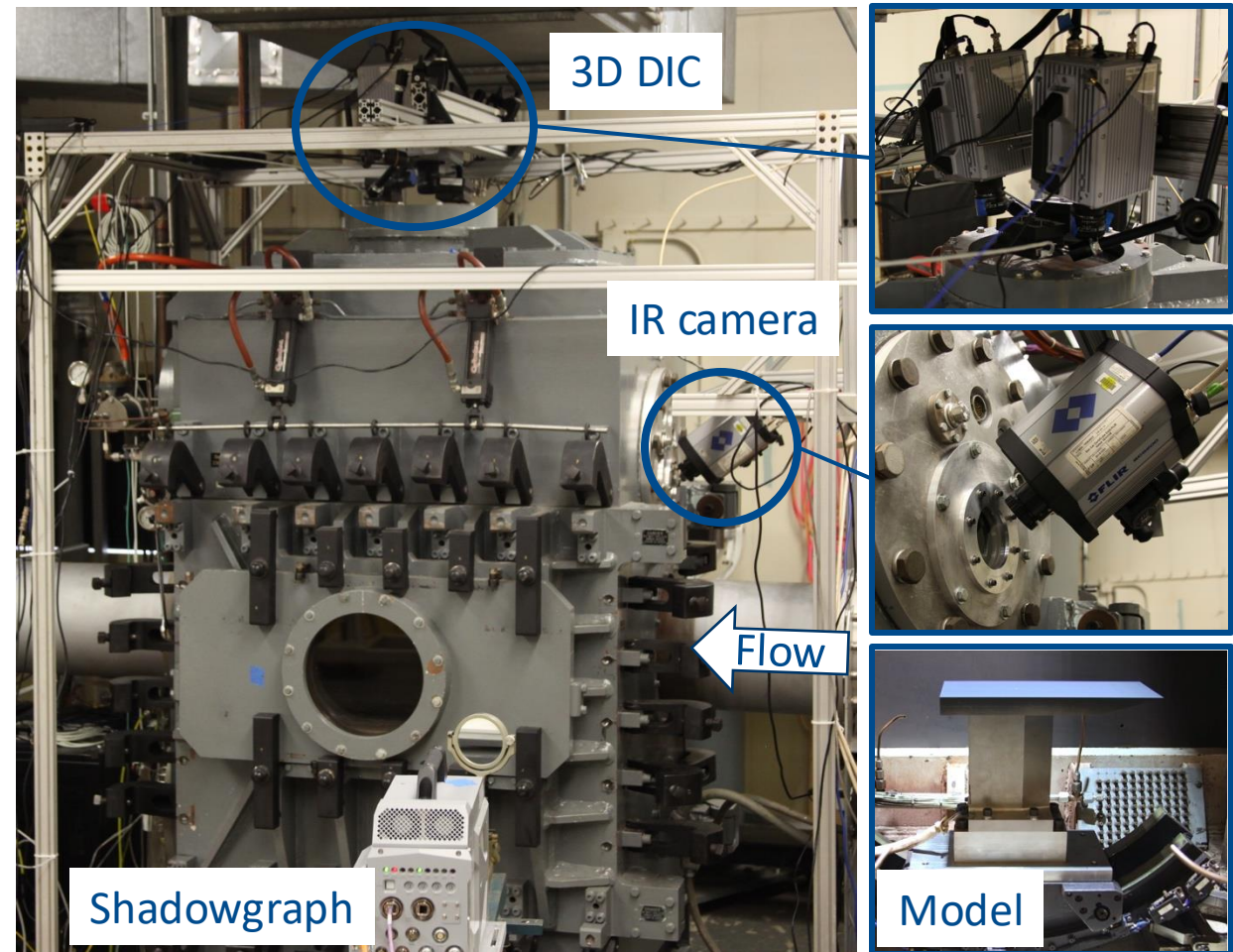
Primary Objective: Measure post-flutter dynamics of thermally buckled panel in M6 flow

M6 High Reynolds Number Facility (M6HRF)

- Free-jet with 254 mm diameter core flow
- Up to ~4 min. of test time (typically 75 sec.)
- Unit Reynolds number: $35 - 91 \times 10^6 \text{ m}^{-1}$
- Total pressure: 4.8 – 12.4 MPa
- Total temperature: 500 – 611 K

Data package:

- Setup/results files packaged consistent with RC-19
- Discrete: TCs, strain gauge, cavity pressure, facility
- Full-field: DIC and IR



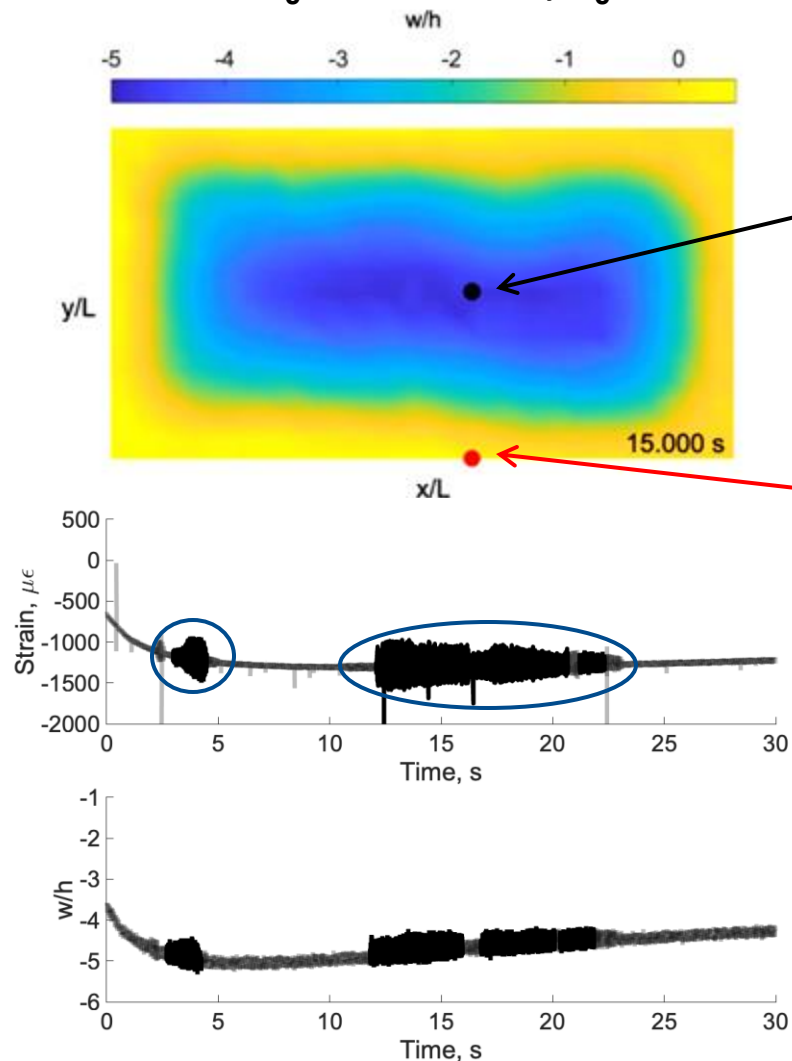


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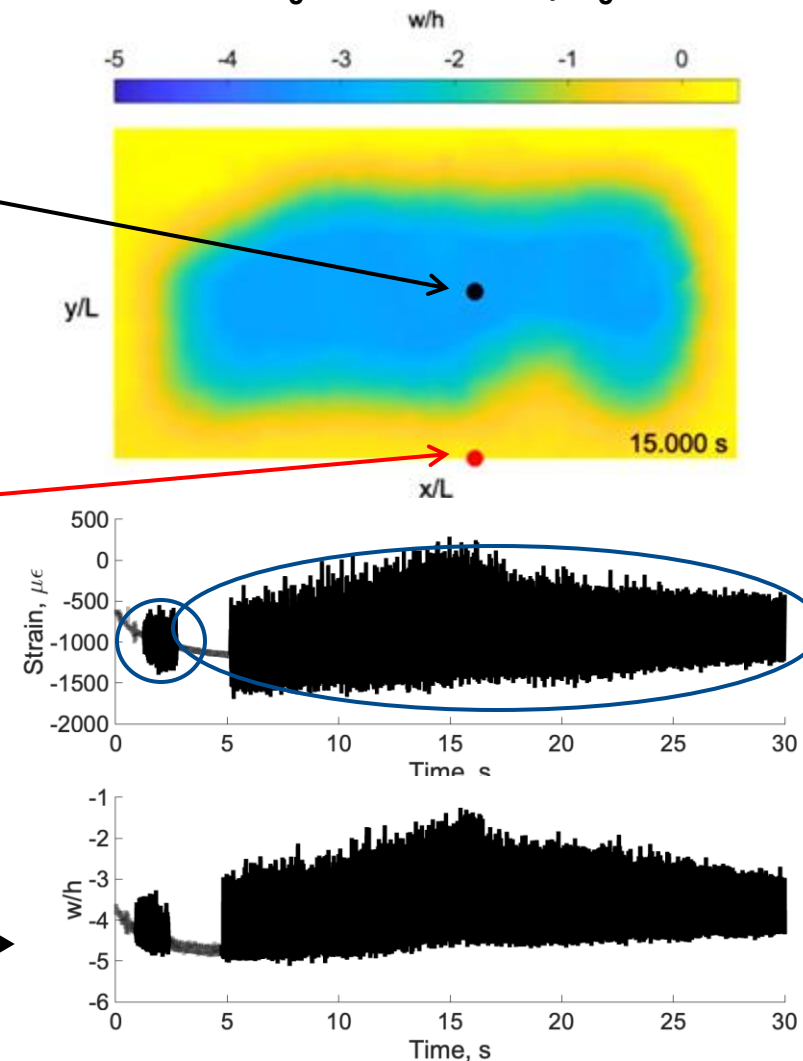


Strain & Displacement Time Histories Highlighting Post-flutter Behavior

Run 29: $P_0 = 12.45$ MPa, $T_0 = 571$ K



Run 31: $P_0 = 12.28$ MPa, $T_0 = 563$ K



DIC

STRAIN

Multiple flutter regions

Increased duration & amplitude for Run 31

HSWG Future Directions: AFRL/DLR H2K Challenge Problem

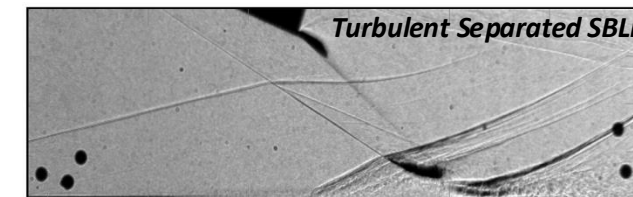
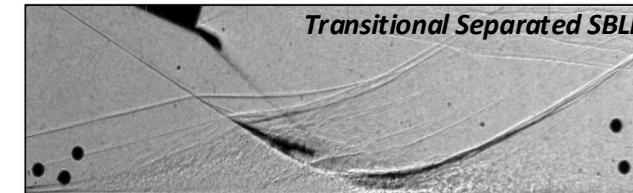
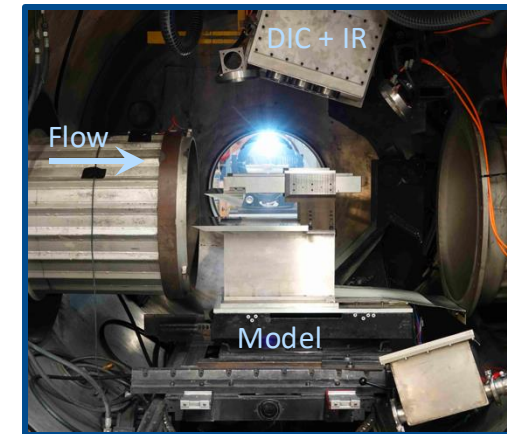
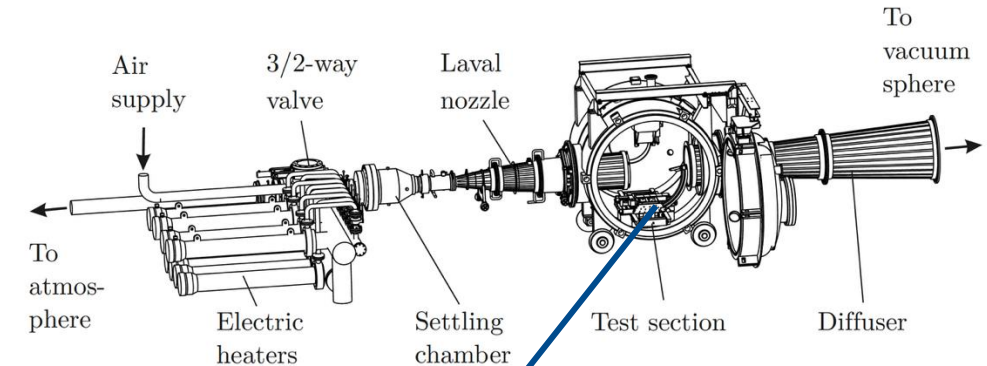
Primary Objective: Measure effects of thermal, pressure, and SBLI loading on AE/ATE instabilities

DLR Hypersonic Wind Tunnel (H2K)

- Free-jet with ~30 sec. of flow time at Mach 5.3
- Unit Reynolds number: $19.3 \times 10^6 \text{ m}^{-1}$
- Total pressure: 1250 kPa
- Total temperature: 390 – 460 K

Data package:

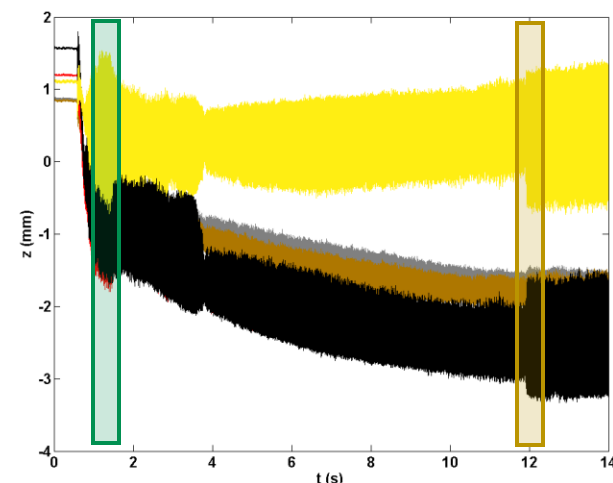
- Working to package setup/results files to be consistent with RC-19 and M6HRF
- Discrete: pressure (rigid only), displacement, cavity pressure, facility
- Full-field: DIC and IR



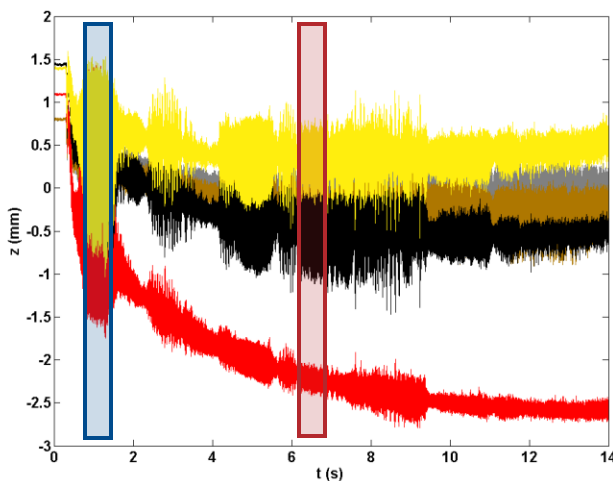
Highlights of Dynamic Instabilities Measured in H2K

Displacement Sensors

20° Shock
 $\Delta x_{Shock} = 0 \text{ cm}$
 $P_0 = 1.24 \text{ MPa}$
 $T_0 = 390 \text{ K}$
 $h_{panel} = 0.7 \text{ mm}$

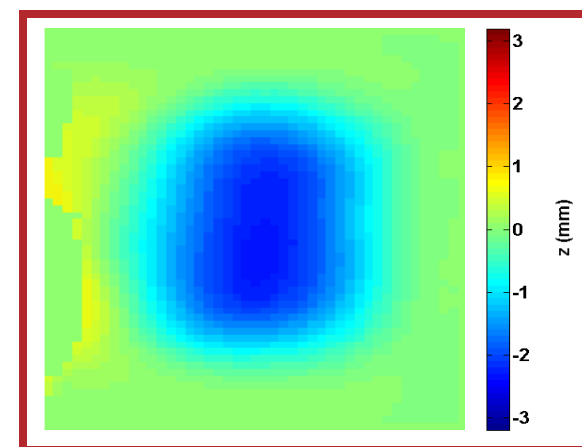
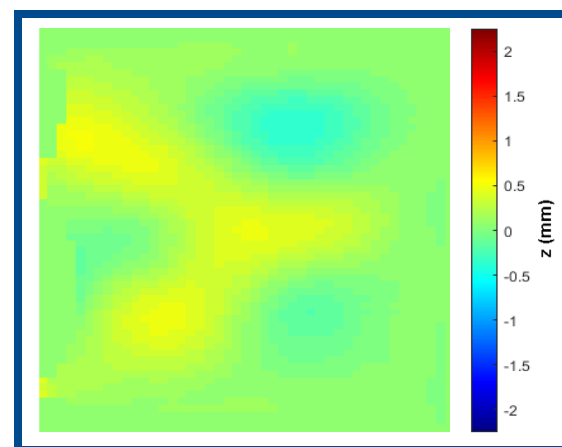
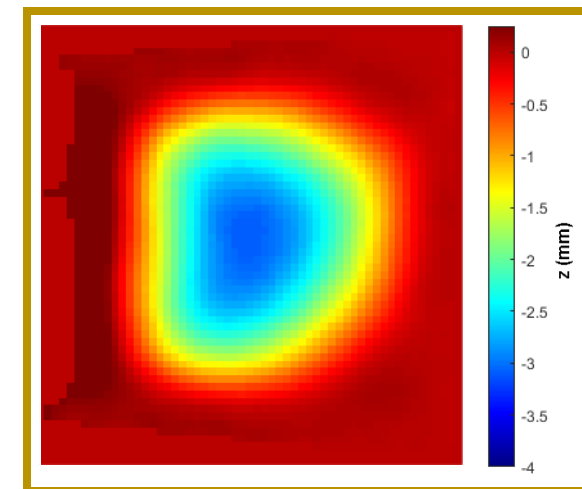
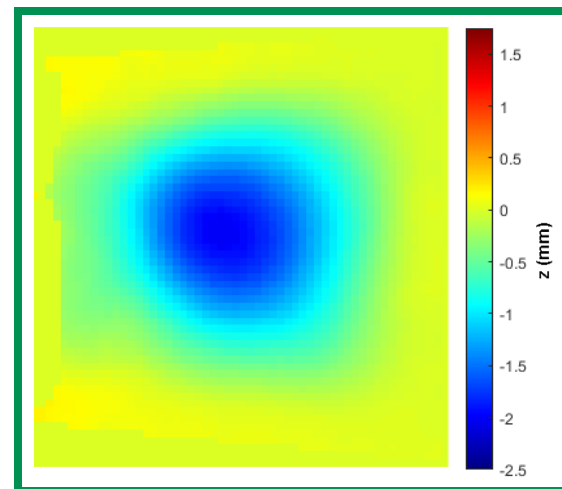


20° Shock
 $\Delta x_{Shock} = 5 \text{ cm}$
 $P_0 = 1.24 \text{ MPa}$
 $T_0 = 410 \text{ K}$
 $h_{panel} = 0.7 \text{ mm}$



— Front
— Center
— Rear
— Left
— Right

DIC at Selected Time Intervals





Questions?